

DATA PREPROCESSING REPORT

Week 2: Data Preprocessing & Feature Engineering

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1. EXECUTIVE SUMMARY

This report documents the comprehensive data preprocessing and feature engineering pipeline applied to the Telco Customer Churn dataset. The raw data has been transformed into a clean, ML-ready dataset through systematic cleaning, feature engineering, encoding, and scaling steps.

2. DATA QUALITY ISSUES IDENTIFIED & RESOLVED

A. Data Type Issues

- Issue: TotalCharges stored as text (object) instead of numeric
- Resolution: Converted to float64 using `pd.to_numeric()`
- Impact: Now usable in calculations and ML models

B. Irrelevant Columns

- Issue: customerID is not predictive
- Resolution: Removed from dataset
- Impact: Reduced noise, improved model focus

C. Missing Values

- Finding: No missing values detected ✓
- Status: No action required

D. Duplicate Records

- Finding: No duplicate records ✓
- Status: No action required

3. FEATURE ENGINEERING APPLIED

TotalServices

Definition: Count of add-on services (OnlineSecurity, OnlineBackup, etc.)

Details: Range: 0-6 services

Rationale: Customers with more services may have stronger loyalty

ContractMonths

Definition: Contract duration converted to months (1, 12, or 24)

Details: Mapping: Month-to-month=1, One year=12, Two year=24

Rationale: Contract length is a strong predictor of churn

ChargePerMonth

Definition: Total charges divided by tenure

Details: Range: Varies by customer

Rationale: Identifies unusual spending patterns

TenureGroup

Definition: Categorical grouping of tenure (0-1 yr, 1-2 yr, 2-4 yr, 4+ yr)

Details: Groups: 4 categories

Rationale: Makes tenure more interpretable for analysis

4. FEATURE ENCODING TECHNIQUES APPLIED

A. Binary Encoding (Yes/No → 1/0)

Applied to: Partner, Dependents, PhoneService, PaperlessBilling, Churn, gender

Rationale: Convert categorical binary values to numeric format for ML algorithms

B. Ordinal Encoding (3-value categories)

Applied to: OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies

Mapping: No=0, No internet service=1, Yes=2

Rationale: Preserves ordinal relationship (no service → limited service → full service)

C. One-Hot Encoding (3+ categories)

Applied to: InternetService, Contract, PaymentMethod

Rationale: Creates separate binary columns for each category

Benefit: Prevents ordinal bias in nominal categorical variables

5. FEATURE SCALING TECHNIQUE APPLIED

Aspect	Details
Method	StandardScaler (Z-score normalization)
Applied to	All numerical features

Formula	$(x - \text{mean}) / \text{standard_deviation}$
Result	Mean = 0, Standard Deviation = 1
Rationale	Improves ML algorithm convergence and performance

6. OUTLIER DETECTION RESULTS

Method: Interquartile Range (IQR)

Decision: Retained all outliers (no removal)

Rationale: Outliers represent real customer behavior and may be predictive of churn

7. VISUALIZATIONS CREATED

- 1. Histograms - Distribution of numerical features
- 2. Box Plots - Outlier detection visualization
- 3. Count Plot - Churn distribution (target variable)
- 4. Correlation Heatmap - Feature relationships with churn
- 5. Scatter Plot - Tenure vs Total Charges (colored by churn status)
- 6. Bar Chart - Churn rate by contract type

8. FINAL DATASET SPECIFICATIONS

Specification	Value
Total Rows	7,043 customers
Total Columns	28 features
Data Types	All numeric (suitable for ML)
Missing Values	0
Duplicate Rows	0
Scaling Applied	Yes (StandardScaler)
Status	✓ READY FOR MODEL DEVELOPMENT

9. KEY FINDINGS & RECOMMENDATIONS

Finding: Contract length is strongly correlated with churn

Recommendation: Use this as a primary feature in the predictive model

Finding: Customers with more services show lower churn rates

Recommendation: Encourage service adoption as a retention strategy

Finding: New customers (low tenure) have highest churn risk

Recommendation: Focus retention efforts on the first year of service

Finding: Data quality is excellent

Recommendation: No additional cleaning needed; dataset is production-ready

10. PREPROCESSING PIPELINE SUMMARY

- ✓ Part 1: Data Inspection - Identified data types, missing values, duplicates
- ✓ Part 2: Data Cleaning - Converted data types, removed irrelevant columns
- ✓ Part 3: Feature Engineering - Created 4 new predictive features
- ✓ Part 4: Feature Encoding & Scaling - Applied encoding and StandardScaler
- ✓ Part 5: Outlier Detection - Identified outliers using IQR method
- ✓ Part 6: Final Dataset - Produced ML-ready dataset

11. CONCLUSION

The Telco Customer Churn dataset has been successfully transformed into a clean, engineered, and ML-ready format. All preprocessing decisions have been documented and justified. The dataset contains 7,043 customers and 20 features, with excellent data quality (0 missing values, 0 duplicates). The dataset is prepared for machine learning model development in Week 3.